Listing 7.50 Creating a custom analyzer for shingles

```
PUT index_with_shingle
{
  "settings": {
    "analysis": {
      "analyzer": {
        "shingles_analyzer":{
          "tokenizer":"standard",
          "filter":["shingles_filter"]
        }
      },
      "filter": {
        "shingles_filter":{
          "type":"shingle",
          "min_shingle_size":2,
          "max_shingle_size":3,
          "output_unigrams":false
        }
      }
    }
  }
}
```

Creates a custom analyzer with a shingles filter

Attributes of the shingle filter (for example, minimum and maximum shingle sizes)

Turns off the output of single words

Invoking this code on some text produces groups of two or three words.

Listing 7.51 Running the `shingle` analyzer

```
POST index_with_shingle/_analyze
{
  "text": "java python go",
  "analyzer": "shingles_analyzer"
}
```

The analyzer returns `[java python, java python go, python go]` because we configured the filter to produce only two- and three-word shingles. Unigrams (one-word shingles) like "java", "python", and so forth are removed from the output because we disabled their output in our filter.

7.8.3 The synonym filter

We worked with synonyms earlier without going into detail. *Synonyms* are different words with the same meanings. For example, if a user searches for "football" or "soccer" (the latter being what football is called in America), either search should point to a football game. The `synonym` filter creates a set of words to produce a richer user experience while searching.

Elasticsearch expects us to provide a set of words and their synonyms by configuring the analyzer with a `synonym` token filter. We create the `synonym` filter on an index's settings as the next listing shows.